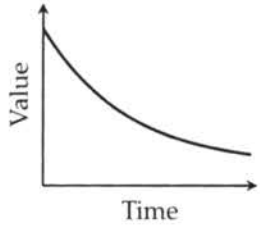
CHAPTER EXERCISE: Answers for this chapter start on page 278.

A calculator should NOT be used on the following questions.

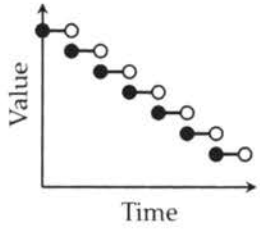
1

The value of a house decreased by 8% from the previous year for n consecutive years. Which of the following graphs could model the value of the house over this time period?

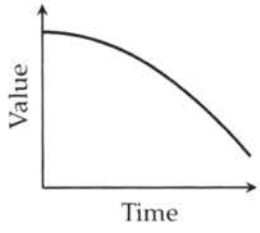
A)



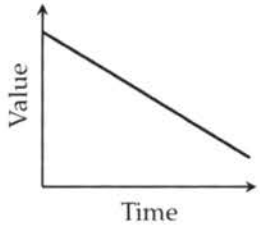
B)



C)



D)



2

The employees at a new bookstore must stock a certain number of shelves so that the store is ready for its opening in two weeks. The employees stock shelves at a constant rate throughout the two weeks. If $p(t)$ is the number of shelves left to be stocked after t days, which of the following statements best describes the function p ?

A) The function p is an increasing exponential function.

B) The function p is a decreasing exponential function.

C) The function p is an increasing linear function.

D) The function p is a decreasing linear function.

3

If the initial population of rats was 20 and grew to 25 after the first year, which of the following functions best models the population of rats P with respect to the number of years t if the population growth of rats is considered to be exponential?

A) $P = 5t + 20$

B) $P = 20(1.25)^t$

C) $P = 20(5)^t$

D) $P = 5t^2 + 20$

4

If the initial population of pandas was 100 and grew to 125 after the first year, which of the following functions best models the population of pandas P with respect to the number of years t if the population growth of pandas is considered to be linear?

A) $P = 25t + 100$

B) $P = 100(1.25)^t$

C) $P = 100(1.2)^t$

D) $P = 20t^2 + 5t + 100$

5

$$f(t) = 20\left(1 + \frac{15}{100}\right)^t$$

The function f above models the temperature, in degrees Celsius, of a metal alloy used in an experiment, where t is the number of seconds after the experiment began. Which of the following is the best interpretation of the number 15 in this context?

A) The temperature, in degrees Celsius, of the metal alloy at the beginning of the experiment

B) The increase in the temperature, in degrees Celsius, of the metal alloy every 100 seconds during the experiment

C) The percent by which the temperature, in degrees Celsius, of the metal alloy decreased from each second to the next during the experiment

D) The percent by which the temperature, in degrees Celsius, of the metal alloy increased from each second to the next during the experiment

6

$$C(t) = 80(2)^{\frac{t}{5}}$$

To examine how a certain virus spreads, scientists introduced the virus to cells in a test tube and found that the number of infected cells in the test tube grew exponentially over time. The function C above models the number of infected cells in the test tube t days after the virus was introduced. Based on the function, which of the following statements is true?

A) The predicted number of infected cells in the test tube doubled every 5 days.

B) The predicted number of infected cells in the test tube grew by a factor of 5 every two days.

C) The predicted number of infected cells in the test tube doubled every day.

D) The predicted number of infected cells in the test tube grew by a factor of 5 every day.

7

$$N = 1{,}000(0.97)^{4h}$$

A scientist uses the equation above to model the number of bacteria N in a petri dish after h hours. According to the model, the number of bacteria is predicted to decrease by 3% every k minutes. What is the value of k ?

A) $\frac{1}{4}$

B) 4

C) 15

D) 240

8

$$C = 4(1.002)^{2t}$$

The equation above can be used to model the number of cars, in millions, registered in a certain state t years after 2009. According to the model, the number of cars registered in the state is projected to increase by n% every 6 months. What is the value of n ?

A) 0.002

B) 0.04

C) 0.2

D) 2

9

The population of trees in a forest has been decreasing by 6 percent every 4 years. The population at the beginning of 2015 was estimated to be 14,000. If P represents the population of trees t years after 2015, which of the following equations gives the population of trees over time?

A) $P = 14{,}000(0.06)^{\frac{t}{4}}$

B) $P = 14{,}000 + 0.94(4t)$

C) $P = 14{,}000(0.94)^{4t}$

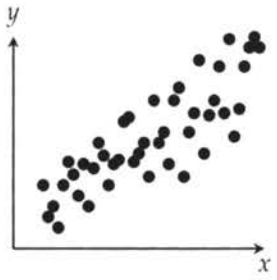
D) $P = 14{,}000(0.94)^{\frac{t}{4}}$

A calculator is allowed on the following questions.

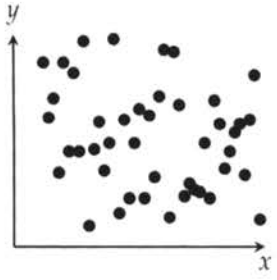
10

Which scatterplot shows the strongest positive association between x and y ?

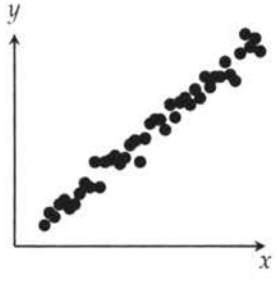
A)



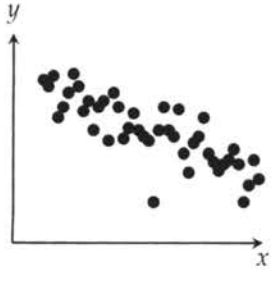
B)



C)



D)



11

Jamie owes Tina some money and decides to pay her back in the following way. Tina receives 3 dollars the first day, 6 dollars the second day, 18 dollars the third day, and 54 dollars the fourth day. Which of the following best describes the relationship between time and the total amount of money (cumulative) Tina has received from Jamie over the course of these four days?

A) Increasing linear
B) Decreasing linear
C) Exponential growth
D) Exponential decay

12

Albert has a large book collection. He decides to trade in two of his used books for one new book each month at a local bookstore. Which of the following best describes the relationship between time (in months) and the total number of books in Albert's collection?

A) Increasing linear
B) Decreasing linear
C) Exponential growth
D) Exponential decay

13

A scientist counts 80 cells in a petri dish and finds that each one splits into two new cells every hour. He uses the function $A(t) = cr^t$ to calculate the total number of cells in the petri dish after t hours. Which of the following assigns the correct values to c and r ?

A) $c = 40, r = 2$
B) $c = 80, r = 0.5$
C) $c = 80, r = 1.5$
D) $c = 80, r = 2$

14

Of the following scenarios, which one would result in linear growth of the square footage of a store?

A) The owner increases the square footage by 0.75% each year.

B) The owner increases the square footage by 5% each year.

C) The owner expands the store by 5% of the original square footage each year.

D) The owner alternates between adding 200 square feet one year and 300 square feet the next year.

15

During its first year of operation, an equipment supplier carried 6,400 items in its product line. For each of the next 6 years, the supplier carried in its product line half the number of items it had carried the previous year. What type of model is best to model the number of items the supplier carried in its product line for any given year in its first 7 years of operation?

A) An exponential growth model

B) An exponential decay model

C) A linear growth model

D) A linear decay model

16

$$V = 200(2^t)$$
$$V = 1,500t$$

An analyst is evaluating how accurate the two models above are in predicting the total number of views, V, an online video receives t days after it is released. How many more views are predicted by the linear model than by the exponential model 4 days after the video is released?

A) 1,400

B) 2,800

C) 3,200

D) 4,000

17

The table below shows the price P, in dollars, of a barrel of crude oil t days after the beginning of an oil shortage.

Number of days, t	Price, P (dollars)
0	50.00
15	60.51
30	73.22
45	88.77

If the equation $P = m(2)^{t/n}$ is used to model the relationship between t and P, which of the following could be the values of m and n ?

A) $m = 25$ and $n = 54.38$

B) $m = 25$ and $n = 86.12$

C) $m = 50$ and $n = 54.38$

D) $m = 50$ and $n = 86.12$

18

The amount of data, in gigabytes, stored in a database increases by 2% every 15 hours. If 16 gigabytes worth of data is currently stored in the database, which of the following functions g gives the amount of data, in gigabytes, that will be stored in the database t days from now?

A) $g(t) = 16(2)^{15t}$

B) $g(t) = 16(1.02)^{\frac{t}{15}}$

C) $g(t) = 16(1.02)^{\frac{5t}{8}}$

D) $g(t) = 16(1.02)^{\frac{8t}{5}}$

Chapter 3: Exponential vs. Linear Growth

CHAPTER EXERCISE:

1. [A] The situation presented in the question is a case of exponential decay. The value of the home decreases more significantly in the beginning and then by smaller and smaller increments over time. Only answer A shows a graph that models exponential decay.

2. [D] Since the employees stock shelves at a constant rate, the number of shelves left to be stocked decreases at a constant rate over time. Therefore, the function p is a decreasing linear function.

3. [B] With exponential growth, we need to calculate the percent increase, which turns out to be $\frac{25-20}{20} = 0.25$. Therefore, the growth factor is 1.25, and the exponential growth can be modeled by $P = 20(1.25)^t$, where 20 is the initial population.

4. [A] The constant increase is $125 - 100 = 25$. Therefore, the slope is 25 and the y-intercept (the initial population) is 100.

5. [D] The given definition of f is in the form of an exponential equation, where 20 is the metal alloy's temperature at the beginning of the experiment and 15 is the percent by which it increased each second. We know the temperature increased each second because the growth factor, $1 + \frac{15}{100} = 1.15$, is greater than 1. For answer C to be correct, the growth factor would have to be $1 - \frac{15}{100} = 0.85$.

6. [A] Since the growth factor is 2 and $\frac{t}{5}$ is the exponent, the predicted number of infected cells doubled every 5 days. So after 5 days, $C(5) = 80(2)^1 = 160$. After 10 days, $C(10) = 80(2)^2 = 320$. And so on.

7. [C] Based on the model, which we can express as $N = 1{,}000(0.97)^{4h} = 1{,}000(0.97)^{h/(1/4)}$, 97% of the bacteria remain after every $\frac{1}{4}$ hour. That's a 3% decrease every $\frac{1}{4} \times 60 = 15$ minutes.

8. [C] Since the growth factor in the equation is 1.002, the number of cars registered increases by $1.002 - 1 = 0.002 = 0.2\%$ every $\frac{1}{2}$ year (6 months). Therefore, $n = 0.2$

9. [D] Since we have a case of exponential decay, we can use the equation $y = ab^{\frac{t}{k}}$ as a reference. The population of trees starts at 14,000 and decays exponentially by a factor of $1 - 0.06 = 0.94$ every 4 years. Therefore, $a = 14000, b = 0.94, k = 4$, and the equation is $P = 14{,}000(0.94)^{\frac{t}{4}}$

10. [C] Scatterplot C is the closest to forming a straight line.

11. [C] Keep track of the total amount she has received: 3, 9, 27, 81. Because the total amount she has received triples each day, the relationship is exponential growth.

12. [B] Each month, Albert loses a book. Because this is a constant decrease, the relationship is linear decay (decreasing linear).

13. [D] The cell count doubles every hour so the growth factor, r, is 2. The initial count is 80 so $c = 80$.

14. $\boxed{C}$ Five percent of the original square footage is a constant. It doesn't change, which would make it linear growth.

15. $\boxed{B}$ Since the number of items gets cut in half every year, the model is one of exponential decay.

16. $\boxed{B}$ The second equation is the linear model: $V = 1,500(4) = 6,000$. The first equation is the exponential model: $V = 200(2^4) = 3,200$. The difference is $6,000 - 3,200 = 2,800$.

17. $\boxed{C}$ The given equation is an exponential growth equation. Since $P = 50$ when $t = 0$ (i.e. at the start), m must equal 50 (you can confirm this by plugging in $t = 0$ and $P = 50$ into the equation). So we know the answer is either choice C or choice D. Now we have to check each of the two answer choices to see which one better approximates the values in the table. If we use a calculator, we'll quickly see that the equation with $n = 54.38$ models the given values of P for $t = 15, 30, 45$ better than the equation with $n = 86.12$.

18. $\boxed{D}$ Let's use the standard exponential equation $y = ab^{\frac{t}{k}}$ as a reference. From the given information, $a = 16$ (the initial amount) and $b = 1.02$ (the growth factor). Since the standard exponential equation requires t and k to be in the same units, we have to convert 15 hours into days, which gives $k = \frac{15}{24} = \frac{5}{8}$ days. Putting everything together, we get $g(t) = 16(1.02)^{t/(5/8)} = 16(1.02)^{\frac{8t}{5}}$.